

Executive Summary: Diabetes Prediction

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Project Goal

This project asks whether self-reported health, lifestyle, and demographic survey indicators can predict whether an individual has diabetes. The model is intended for **screening support**, not clinical diagnosis. In this setting, the most important goal is to identify as many true diabetes cases as possible so that people can be referred for follow-up testing or care.

The raw target variable, `Diabetes_012`, has three classes:

Original label	Meaning
0	No diabetes
1	Prediabetes
2	Diabetes

For the current modeling pipeline, we converted this into a binary target:

Binary label	Meaning
0	No diabetes or prediabetes
1	Diabetes

This binary framing focuses the project on identifying respondents who already report diabetes.

Data Processing and Duplicate Decision

The raw BRFSS-derived dataset contains 253,680 rows and 22 columns, including the target. During the initial cleaning stage, we removed exact duplicate rows before converting the target to binary. This removed 23,899 rows, or about 9.42% of the raw data.

The original reasoning was that each row represents one individual respondent, and it seemed unlikely that two independent people would share the exact same values across all 22 variables, including the diabetes target. Under that assumption, exact duplicate rows could represent repeated records and could overweight common profiles if left in the dataset.

However, 9.42% is a large share of the dataset. Also, this survey dataset contains many discrete and binary variables, so two different people can realistically have the same feature profile. For that reason, we treated duplicate removal as an important modeling assumption and later performed a sensitivity check.

Duplicate Sensitivity Check

The sensitivity check is documented in:

`notebooks/duplicate_sensitivity_check.ipynb`

The notebook compares two scenarios:

Scenario	Description
Drop exact duplicates	Main project pipeline
Keep exact duplicates	Sensitivity-check pipeline

Duplicate diagnostics showed:

Check	Result
Exact duplicate rows including target	23,899
Exact duplicate rows as percent of raw data	9.42%
Exact duplicate groups including target	11,187
Largest exact duplicate group	59
Feature duplicate rows ignoring target	25,772
Feature duplicate rows as percent of raw data	10.16%
Feature duplicate groups with conflicting original target labels	1,834

The conflicting feature-only duplicate groups are important. They show that respondents can have exactly the same predictor values but different original diabetes labels. This supports the idea that not all duplicate-looking feature profiles are data errors. In a survey dataset with mostly categorical features, repeated profiles can occur naturally.

The sensitivity check reruns the full modeling workflow under both duplicate-handling choices:

1. Convert `Diabetes_012` to binary `Diabetes_01`.
2. Create a stratified 80/20 train-test split.
3. Use 5-fold stratified cross-validation on the training set.
4. Select thresholds from train-only out-of-fold predictions.
5. Refit each model on the full training set.
6. Evaluate on the held-out test set.

The sensitivity check does **not** perform new hyperparameter tuning. It uses the same model settings from the main model-selection notebooks so that the comparison isolates the duplicate-removal decision.

Overall, keeping duplicates changed some metrics, especially for the neural network and several high-recall threshold settings, but it did not overturn the broad project conclusion: the survey indicators contain meaningful predictive signal, XGBoost remains one of the strongest ranking models, and logistic regression remains a competitive interpretable benchmark. Therefore, removing exact duplicates is a defensible primary analysis choice, but the final report should clearly disclose that the result was checked against a keep-duplicates sensitivity scenario.

Modeling Strategy

The cleaned binary dataset was split into stratified train and test sets with `random_state=42`. Model selection used 5-fold stratified cross-validation on the training set. Final testing was performed separately in:

`notebooks/all_model_testing.ipynb`

The models evaluated were:

Model family	Notes
Dummy classifier	Majority-class baseline

Model family	Notes
Logistic regression	Regular logistic, Lasso logistic, and Ridge logistic
Random forest	Class-weighted random forest
XGBoost	Gradient-boosted tree classifier
Support vector machine	Standardized features with class weighting
MLP neural network	Keras feed-forward neural network

Because this is a screening task, the primary metric is **recall** for the diabetes class. Secondary metrics include precision, F1, F2, AUPRC, AUROC, predicted positive rate, and $\text{TPR} - \text{FPR}$.

Threshold Strategy

For each non-baseline model, we compared three threshold strategies:

Threshold strategy	Meaning
Default threshold	0.5 for probability models and 0.0 for SVM decision scores
Max F2 threshold	Chooses the threshold that maximizes F2, emphasizing recall
Max TPR-FPR threshold	Chooses the ROC operating point that maximizes $\text{TPR} - \text{FPR}$

Threshold choice is not only a technical decision. It is also a public health, medical, or business operations decision. A lower threshold increases recall but flags more people for follow-up, which lowers precision and increases workload.

Final Test-Set Findings

On the final held-out test set, the dummy classifier had high apparent accuracy because the data are imbalanced, but it failed to identify diabetes cases. This confirms that accuracy alone is not a useful model-selection metric for this project.

The strongest results were generally produced by XGBoost, the neural network, SVM, and logistic regression depending on which metric was prioritized.

Key findings:

Finding	Interpretation
Neural network had the highest test recall under the max-F2 threshold	It found the largest share of diabetes cases, but at the cost of many false positives.
XGBoost had the strongest ranking metrics	It achieved the best AUPRC and AUROC among the tested models.
Logistic regression was surprisingly competitive	Regular, Lasso, and Ridge logistic regression behaved very similarly.
SVM was close to logistic regression	This suggests that much of the learnable signal is approximately linear in the current feature set.
Random forest was more conservative at the default threshold	It predicted fewer positives, giving lower recall but sometimes higher precision.

The top models by test recall were all using the max-F2 threshold. This is expected because max-F2 explicitly favors recall.

Approximate top recall results from the final test evaluation:

Rank	Model	Threshold	Recall	Precision	F2	AUPRC	AUROC
1	MLP neural network	Max F2	0.864	0.280	0.609	0.437	0.817
2	SVM	Max F2	0.857	0.284	0.611	0.414	0.811
3	Ridge logistic regression	Max F2	0.856	0.285	0.611	0.413	0.812
4	Regular logistic regression	Max F2	0.856	0.285	0.611	0.413	0.812
5	Lasso logistic regression	Max F2	0.856	0.285	0.611	0.413	0.812

XGBoost did not have the highest recall in this ranking, but it had the strongest ranking quality:

Model	Test AUPRC	Test AUROC
XGBoost	0.443	0.820
MLP neural network	0.437	0.817
Logistic regression / SVM group	about 0.413-0.414	about 0.811

This creates a useful final interpretation: if the goal is the best overall ranking model, XGBoost is the strongest candidate. If the goal is maximum recall at a chosen operating point, the neural network and logistic regression and SVM are also highly competitive.

Logistic Regression and Interpretability

Regular logistic regression, Lasso, and Ridge produced very similar results. This likely happened because:

1. The dataset is large, so regular logistic regression is already stable.
2. The number of features is moderate.
3. The model is not severely over-parameterized.
4. Regularization changes coefficient size but does not greatly change the feature-importance ordering.

This is useful because logistic regression provides an interpretable benchmark. The important logistic coefficients broadly align with the EDA heatmap: features such as general health, high blood pressure, BMI, age, high cholesterol, and difficulty walking are meaningfully related to diabetes status.

Implications

The project shows that self-reported BRFSS health indicators can meaningfully classify diabetes status better than a dummy baseline. However, even the stronger models have modest precision when recall is high. This is not surprising because the positive diabetes class is much smaller than the negative class.

For a screening workflow, this tradeoff may be acceptable. The model should not be used to diagnose diabetes, but it could help identify people who may benefit from confirmatory clinical testing or preventive follow-up.

A practical recommendation would be:

Use case	Recommended model direction
Highest recall screening	Neural network or linear model with max-F2 threshold
Best predictive ranking	XGBoost
Most interpretable model	Logistic regression

Limitations

Important limitations remain:

1. The data are self-reported survey responses, so they may contain recall bias and reporting error.
2. The model predicts reported diabetes status, not clinically measured diabetes.
3. Prediabetes is grouped with non-diabetes in the current binary target.
4. The duplicate-removal decision affects dataset size and should be disclosed.
5. The data are from BRFSS 2015, so relationships may differ in newer populations.
6. The model should not be interpreted as causal.

Future Work

1. Three-Class Diabetes Modeling

The current project converts `Diabetes_012` into binary `Diabetes_01`. A natural next step is to keep all three original classes:

Class	Meaning
0	No diabetes
1	Prediabetes
2	Diabetes

For neural networks, this would use a softmax output layer with three units and a multiclass loss such as sparse categorical cross-entropy. For other models, multiclass logistic regression, XGBoost multiclass classification, and random forest multiclass classification could be compared.

This would be clinically and operationally useful because prediabetes is an important prevention stage rather than simply part of the negative class.

2. Better Threshold and Cost Analysis

The current project compares default, max-F2, and max-TPR-FPR thresholds. Future work should choose thresholds using explicit operational costs:

Cost consideration	Example
False negative cost	Missed diabetes case
False positive cost	Extra follow-up screening
Capacity constraint	Number of people clinics can follow up with
Equity constraint	Whether errors differ across demographic groups

This would make threshold selection more realistic than choosing a threshold from a metric alone.

3. Model Improvement and External Validation

Future work should validate the model on newer BRFSS data or another external dataset. Additional work could include probability calibration, subgroup fairness analysis, feature engineering, and comparison with clinically measured variables if available.

External validation is especially important before using any model in a real screening or public health setting.